

**ASSIGNMENT No: 17****TITLE: Write a Java program to demonstrate the use of Swing and Applets**

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

Theory

Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.

Code :

```
import java.lang.*;

import javax.swing.*;

public class SwingDemo {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Swing Demo");

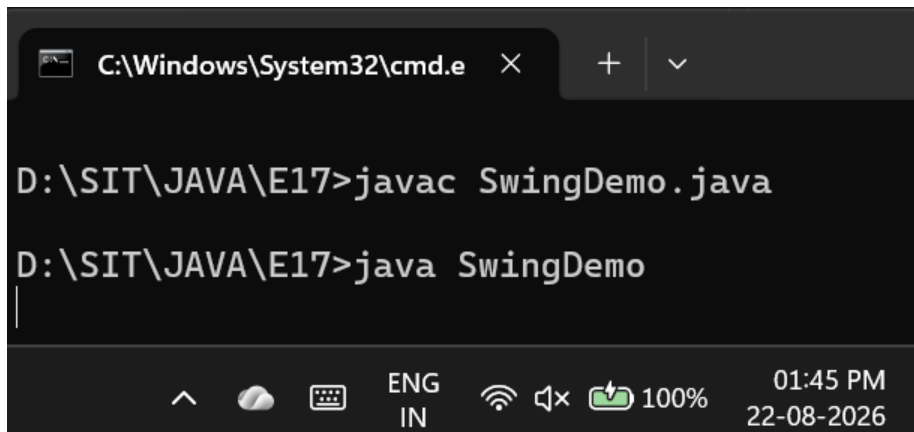
        JLabel label = new JLabel("Welcome to Swing!", SwingConstants.CENTER);

        frame.add(label);

        frame.setSize(400, 300);

        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

Output:

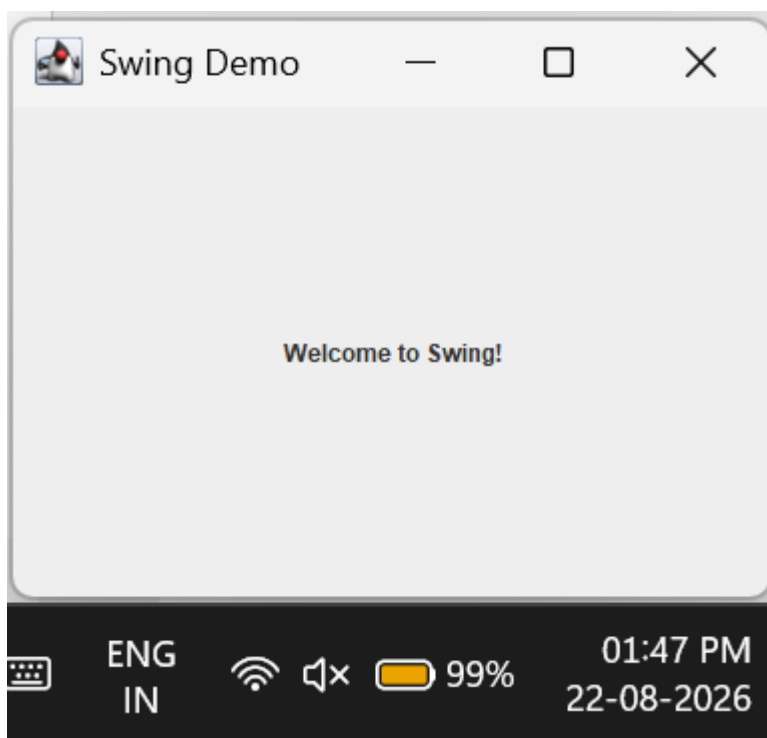


```
C:\Windows\System32\cmd.e X + v

D:\SIT\JAVA\E17>javac SwingDemo.java

D:\SIT\JAVA\E17>java SwingDemo
|
```

The screenshot shows a Windows command prompt window with a dark background. The title bar indicates the path 'C:\Windows\System32\cmd.e'. The command prompt shows the user navigating to 'D:\SIT\JAVA\E17' and running 'javac SwingDemo.java' followed by 'java SwingDemo'. The system tray at the bottom shows the time as 01:45 PM on 22-08-2026, along with icons for network, volume, and battery (100%).



Code :

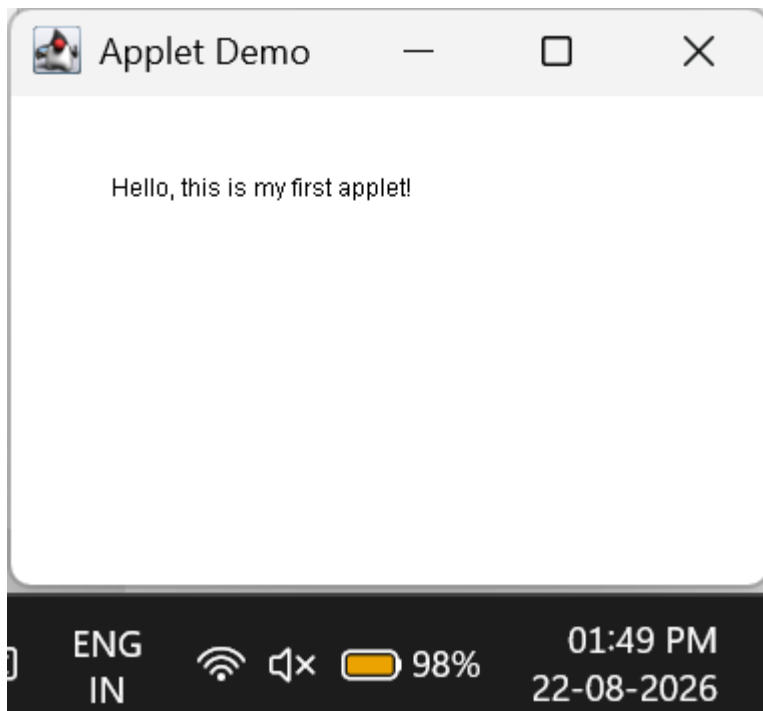
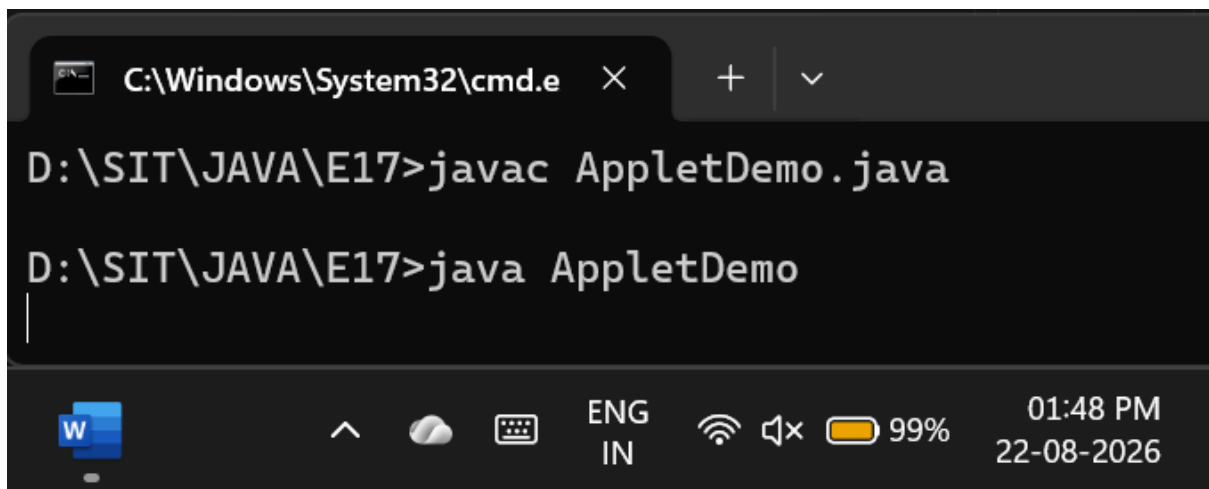
```
import java.lang.*;

import java.applet.Applet;
import java.awt.*;

public class AppletDemo extends Applet {
    public void paint(Graphics g) {
        g.drawString("Hello, this is my first applet!", 50, 50);
    }
}
```

```
public static void main(String[] args) {  
    Frame frame = new Frame("Applet Demo");  
    AppletDemo applet = new AppletDemo();  
    frame.add(applet);  
    frame.setSize(400, 300);  
    frame.setVisible(true);  
}  
}
```

Output:



Exercise

1. Create a simple **Student Registration Form GUI** using Swing.

Code :

```
package E17;

import java.lang.*;
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;

public class StudentRegistrationForm {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Student Registration");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setSize(450, 350);
        frame.setLocationRelativeTo(null);

        JPanel panel = new JPanel();
        panel.setLayout(new GridLayout(5, 2, 15, 15));
        panel.setBorder(new EmptyBorder(25, 25, 25, 25));
        panel.setBackground(new Color(240, 248, 255));

        Font font = new Font("Arial", Font.BOLD, 14);

        JLabel nameLabel = new JLabel("Student Name:");
        nameLabel.setFont(font);
        JTextField nameField = new JTextField();

        JLabel rollLabel = new JLabel("Roll Number:");
        rollLabel.setFont(font);
        JTextField rollField = new JTextField();

        JLabel courseLabel = new JLabel("Course:");
        courseLabel.setFont(font);
        JTextField courseField = new JTextField();

        JButton submitBtn = new JButton("Submit");
        submitBtn.setBackground(new Color(70, 130, 180));
```

```
submitBtn.setForeground(Color.WHITE);
submitBtn.setFont(font);
submitBtn.setFocusPainted(false);
```

```
JButton clearBtn = new JButton("Clear");
clearBtn.setBackground(new Color(220, 53, 69));
clearBtn.setForeground(Color.WHITE);
clearBtn.setFont(font);
clearBtn.setFocusPainted(false);
```

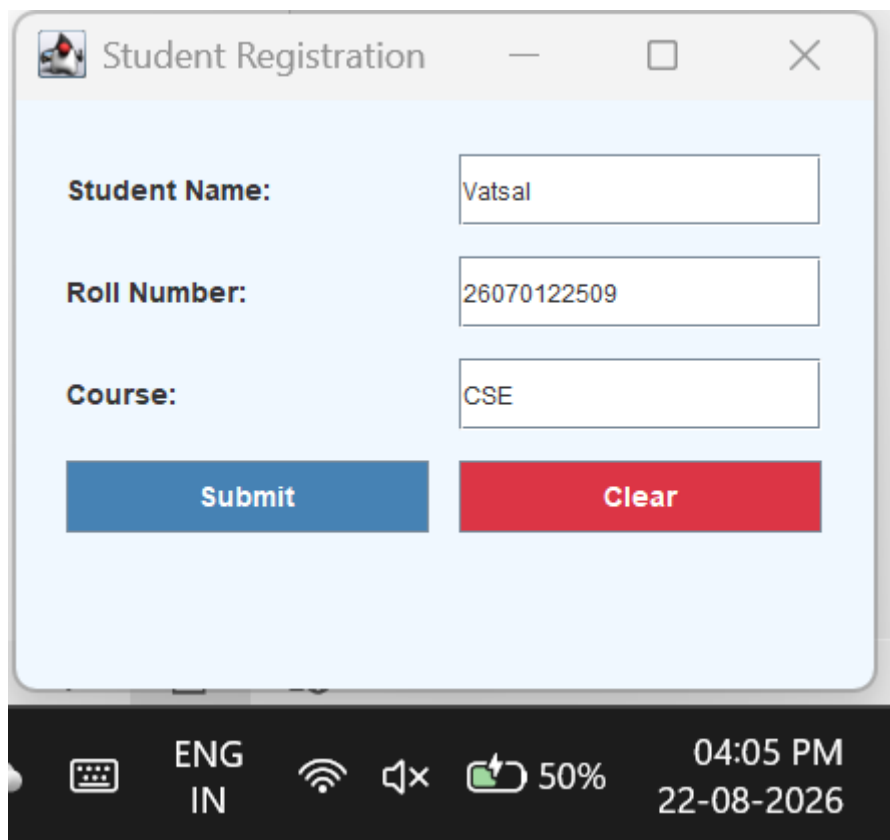
```
panel.add(nameLabel);
panel.add(nameField);
panel.add(rollLabel);
panel.add(rollField);
panel.add(courseLabel);
panel.add(courseField);
panel.add(submitBtn);
panel.add(clearBtn);
```

```
submitBtn.addActionListener(e -> {
    String msg = "Student Registered Successfully!\n\nName: " + nameField.getText() +
        "\nRoll No: " + rollField.getText() +
        "\nCourse: " + courseField.getText();
    JOptionPane.showMessageDialog(frame, msg, "Success",
JOptionPane.INFORMATION_MESSAGE);
});
```

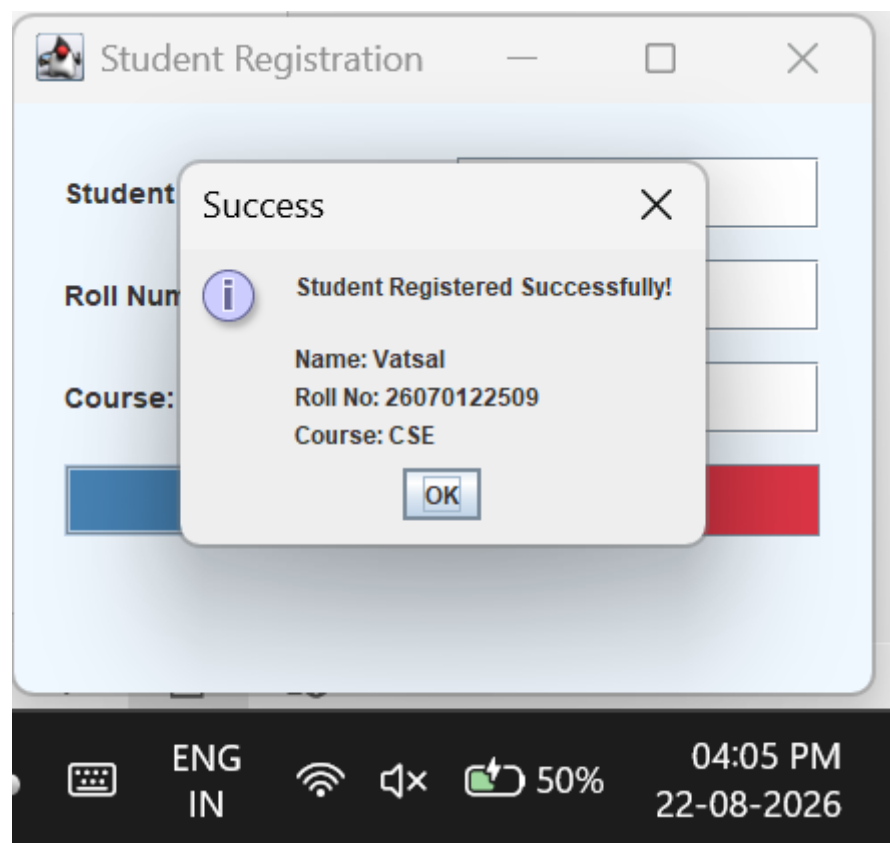
```
clearBtn.addActionListener(e -> {
    nameField.setText("");
    rollField.setText("");
    courseField.setText("");
});
```

```
frame.add(panel);
frame.setVisible(true);
}
}
```

Output:



The screenshot shows a window titled "Student Registration" with a light blue background. It contains three input fields: "Student Name:" with the value "Vatsal", "Roll Number:" with the value "26070122509", and "Course:" with the value "CSE". Below the fields are two buttons: a blue "Submit" button and a red "Clear" button. The window has standard OS controls (minimize, maximize, close) in the title bar. At the bottom, a black status bar displays icons for keyboard, network, and battery, along with the text "ENG IN", "50%", and the time/date "04:05 PM 22-08-2026".



This screenshot shows the same "Student Registration" window, but with a "Success" dialog box overlaid in the center. The dialog box has a title bar with a close button and an information icon. The message inside reads "Student Registered Successfully!". Below this, it lists the registered details: "Name: Vatsal", "Roll No: 26070122509", and "Course: CSE". An "OK" button is at the bottom of the dialog. The background window is dimmed. The status bar at the bottom remains the same, showing "ENG IN", "50%", and "04:05 PM 22-08-2026".

2. Create a simple **Employee Registration Form** using **Java Swing GUI** that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box

Code :

```
package E17;

import java.lang.*;
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;

public class EmployeeRegistrationForm {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Employee Registration");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setSize(450, 400);
        frame.setLocationRelativeTo(null);

        JPanel panel = new JPanel();
        panel.setLayout(new GridLayout(6, 2, 15, 15));
        panel.setBorder(new EmptyBorder(25, 25, 25, 25));
        panel.setBackground(new Color(245, 245, 245));

        Font font = new Font("Segoe UI", Font.BOLD, 14);

        JLabel idLabel = new JLabel("Employee ID:");
        idLabel.setFont(font);
        JTextField idField = new JTextField();

        JLabel nameLabel = new JLabel("Employee Name:");
        nameLabel.setFont(font);
        JTextField nameField = new JTextField();

        JLabel deptLabel = new JLabel("Department:");
        deptLabel.setFont(font);
        JTextField deptField = new JTextField();
    }
}
```

```

JLabel salaryLabel = new JLabel("Salary:");
salaryLabel.setFont(font);
JTextField salaryField = new JTextField();

JButton submitBtn = new JButton("Register");
submitBtn.setBackground(new Color(40, 167, 69));
submitBtn.setForeground(Color.WHITE);
submitBtn.setFont(font);
submitBtn.setFocusPainted(false);

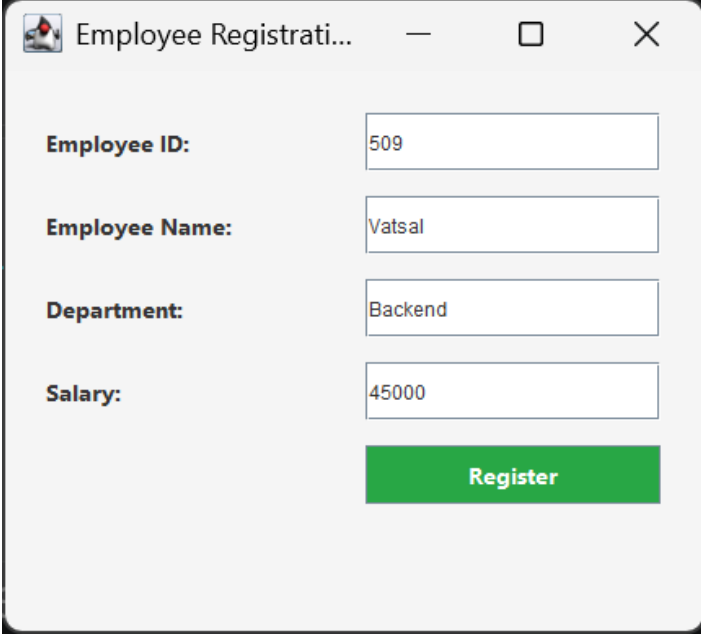
panel.add(idLabel);
panel.add(idField);
panel.add(nameLabel);
panel.add(nameField);
panel.add(deptLabel);
panel.add(deptField);
panel.add(salaryLabel);
panel.add(salaryField);
panel.add(new JLabel(""));
panel.add(submitBtn);

submitBtn.addActionListener(e -> {
    String msg = "--- Employee Registration Successful ---\n\n" +
        "Employee ID: " + idField.getText() + "\n" +
        "Name: " + nameField.getText() + "\n" +
        "Department: " + deptField.getText() + "\n" +
        "Salary: " + salaryField.getText();
    JOptionPane.showMessageDialog(frame, msg, "Registration Info",
JOptionPane.INFORMATION_MESSAGE);
});

frame.add(panel);
frame.setVisible(true);
}
}

```

Output:



Employee Registrati...

Employee ID: 509

Employee Name: Vatsal

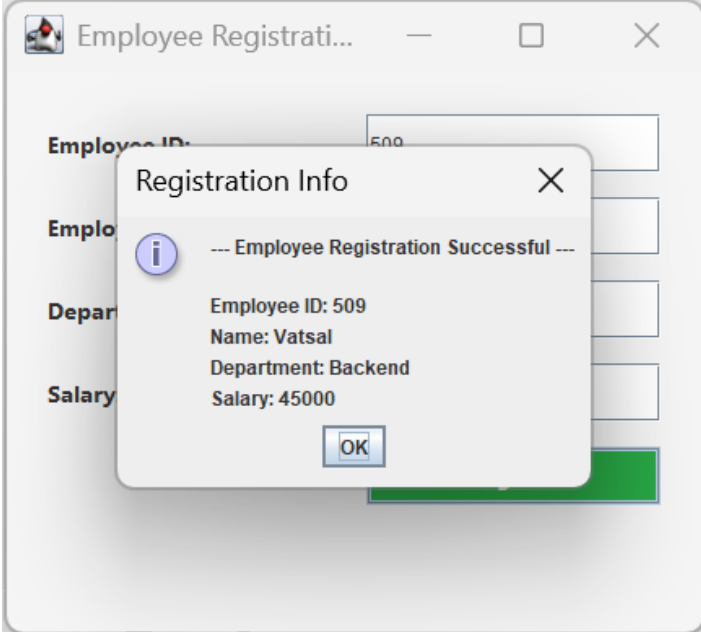
Department: Backend

Salary: 45000

Register

ENG IN 04:16 PM 22-08-2026

The image shows a mobile application window titled "Employee Registrati...". It contains four input fields: "Employee ID" with the value "509", "Employee Name" with "Vatsal", "Department" with "Backend", and "Salary" with "45000". Below these fields is a green button labeled "Register". At the bottom of the screen is a status bar with icons for keyboard, language (ENG IN), Wi-Fi, signal, and battery (55%), along with the time "04:16 PM" and date "22-08-2026".



Registration Info

--- Employee Registration Successful ---

Employee ID: 509
Name: Vatsal
Department: Backend
Salary: 45000

OK

Employee ID: 509

Employee Name: Vatsal

Department: Backend

Salary: 45000

ENG IN 04:16 PM 22-08-2026

The image shows the same "Employee Registrati..." window as above, but with a "Registration Info" dialog box overlaid in the center. The dialog box has a title bar with a close button (X) and an information icon (i). The message inside says "--- Employee Registration Successful ---". Below this, it lists the registration details: "Employee ID: 509", "Name: Vatsal", "Department: Backend", and "Salary: 45000". At the bottom of the dialog is an "OK" button. The background window is slightly dimmed. The status bar at the bottom remains the same.

Conclusion:

I successfully demonstrated the creation of graphical user interfaces using Java Swing and Applets. We learned how to construct functional applications using lightweight Swing components—such as frames, labels, text fields, and buttons—while utilizing layout managers to organize them neatly. Additionally, we implemented basic event handling to capture user input and make the applications interactive.